inspection in New York and California

In the United States, UAVs have become essential tools for monitoring and maintaining critical urban and industrial infrastructure, particularly in states such as New York and California, where aging networks and complex terrain present challenges for conventional inspection methods. Public agencies and utility companies now deploy drones to inspect bridges, power transmission lines, pipelines, rail infrastructure, and high-rise building façades, replacing traditional inspection processes that often require scaffolding, helicopter flyovers, or manual climbing. These drone-based inspection systems significantly reduce operational risk, cost, and inspection time, while produc ing high-resolution, geo-referenced imagery and structural condition datasets that support predictive maintenance.

In New York, the New York State Department of Transportation (NYSDOT) uses UAVs equipped with LiDAR scanners and optical zoom cameras to evaluate bridge joints, steel truss integrity, and corrosion hotspots in complex structural segments that are difficult or dangerous to examine manually. This approach eliminates the need to close roadways or erect temporary platforms, allowing inspections to be completed safely and with minimal interruption to urban mobility. UAV-generated digital models are incorporated into building information modeling (BIM) systems and digital twin platforms, enabling long-term structural monitoring and lifecycle planning (Duque, et al. 2018).

In California, the energy utility company Pacific Gas & Electric (PG&E) utilizes UAVs to inspect thousands of kilometers of overhead power lines that traverse mountainous, forested, and wildfire-prone regions. Equipped with thermal cameras and vegetation encroachment detection algorithms, these drones help identify frayed lines, overheated transformers, and hazardous tree overgrowth conditions. By detecting risks early, UAV-based